DEPARTMENT OF PROFESSIONAL EDUCATION

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ASSIGNMENT REPORT

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ELECTRONIC INDUSTRIAL AUTOMATION

INDUSTRY AUTOMATION II

TITLE: **DESIGN AND SIMULATION OF LADDER DIAGRAM PROGRAMMING USING TIA PORTAL AND FACTORY IO**

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ABSTRACT

The design and modeling of an automated sorting station utilizing Siemens TIA Portal and Factory IO is the main goal of this assignment. Conveyors and sensors in a virtual industrial setting are controlled by a ladder diagram (LAD) application. Workpieces were detected and sorted using Boolean-based sensors as inputs, and smooth and controlled operation was ensured by the application of timers and counters. Factory IO simulation was used to test the system, and proper sequence execution and sorting behavior were noted. Due to a lack of information and inadequate understanding of processing non-Boolean data in ladder logic programming, RFID technology was not used. All things considered, this project illustrates the basic ideas of virtual commissioning and PLC programming for industrial automation applications.

Keyword: *PLC, Ladder Diagram, TIA Portal, Factory IO, Industrial Automation*

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INTRODUCTION

A. Background

Modern production depends heavily on industrial automation because it increases productivity, precision, and operational safety. Because of their resilience and capacity to link with a wide range of sensors and actuators, programmable logic controllers, or PLCs, are frequently utilized as the primary control devices in automated systems. Ladder Diagram (LAD) is supported by the well-known PLC programming framework Siemens TIA Portal, which makes it appropriate for both industrial and educational uses.

Additionally, automation systems can be built and evaluated in a virtual environment thanks to simulation tools like Factory IO. Without the requirement for actual hardware, students can test control logic, visualize system behavior, and find faults by combining PLC programs with Factory IO. This method lowers development risks and improves comprehension of automation principle

B. Purpose of Research

This assignment's goal is to develop and simulate an automated sorting station utilizing Factory IO as the hardware simulation platform and TIA Portal for ladder diagram programming. This study focuses on controlling conveyor-based material handling processes using basic PLC programming components like timers, counters, and Boolean logic. Evaluating design choices made during system development, especially the selection of sensor technologies, is another goal of the assignment.

C. Discovery Question

Below is the discovery question in this study:

1. *How can timers be implemented and used in TIA Portal to control delays and sequencing in ladder diagram programming?*

2. *How can counters be applied in TIA Portal to count and monitor the number of workpieces in an automated sorting system?*

3. *Why is RFID technology not implemented in this project, especially in relation to non-Boolean data handling and programming limitations?*

METHODS

A. Overviews

This assignment combines industrial automation software tools and Programmable Logic Controller (PLC) programming to create and simulate an automated sorting system. Siemens TIA Portal with Ladder Diagram (LAD) programming, which is frequently used in industrial settings because of its ease of use and resemblance to electrical relay circuits, is used to develop and implement the system's control logic. To guarantee precise and sequential system operation, the sorting process is managed by Boolean sensor signals, timers, and counters.

To simulate the physical behavior of the automation system, Factory IO is used as a virtual hardware platform. Conveyor belts, sensors, actuators, and other real-world components are represented in a three-dimensional industrial environment provided by Factory IO. This lowers costs and increases safety during development by enabling the sorting station's behavior to be seen and tested without the requirement for real hardware.

Communication between TIA Portal and Factory IO is achieved using Siemens PLCSIM, which acts as a virtual PLC. Real-time data transmission between the PLC logic and the Factory IO simulation is made possible by downloading and running the ladder program developed in TIA Portal in PLCSIM. The PLC receives sensor signals produced in Factory IO as input signals, and the PLC's output signals regulate the motion and operations of the system's simulated actuators.

All things considered, the combination of Factory IO, LAD programming, PLCSIM, and TIA Portal creates a comprehensive virtual commissioning environment. With this method, industrial automation principles, control logic design, and system integration can be practically understood while the PLC program can be effectively tested, validated, and enhanced.

B. Hardware Simulation

The sorting station's operational environment and physical components are represented in a virtual three-dimensional world using Factory IO as the hardware simulation platform. Important industrial components including conveyor belts for material movement, sensors for workpiece presence and location, and diverter routes that regulate each item's sorting orientation are all modeled in the simulation. Realistic system behavior can be seen because these parts are set up to closely mimic an actual industrial sorting system.

Real-time connection between the control logic and the simulated hardware is accomplished by combining Factory IO with the PLC program created in TIA Portal. Conveyors and sorting mechanisms in the simulation are controlled by the PLC's output instructions, which receive input data from sensors in Factory IO. Continuous system response monitoring and program execution verification are made possible by this real-time interaction.

Factory IO reduces cost, setup time, and potential safety issues by doing away with the requirement for real hardware throughout the development and testing phases. Additionally, it offers an adaptable learning environment that makes fault testing, logic modifications, and design changes simple. Because of this, Factory IO is a useful tool for verifying PLC control logic and comprehending automated system behavior prior to actual installation.

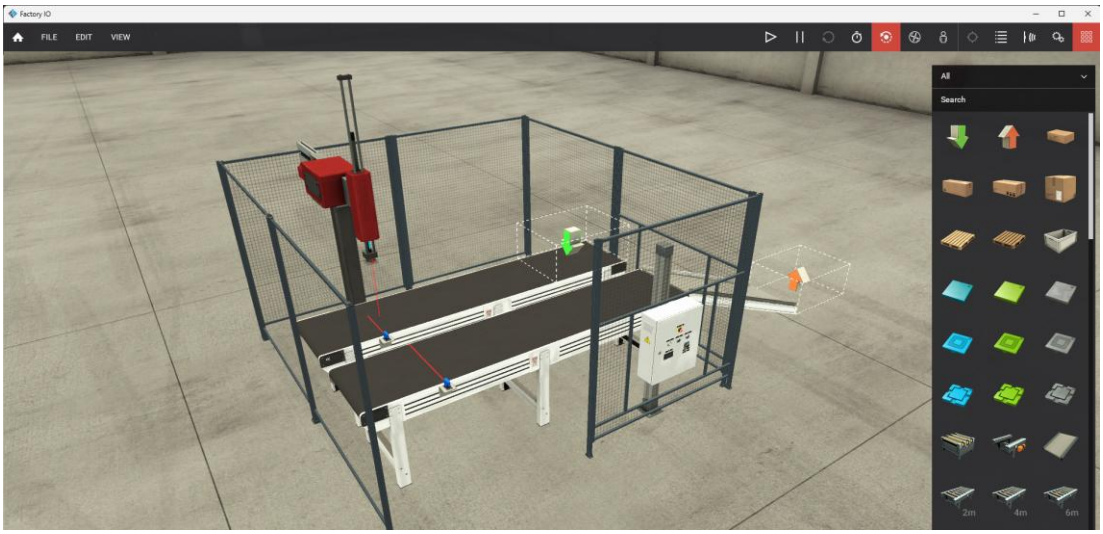


Figure 1. Factory IO

- **Materials, Equipment and Components Used in Simulation**

These materials were intentionally selected to demonstrate the system's capacity to discriminate between material composition (Plastic vs. Metal) and shape (Lid vs. Base).

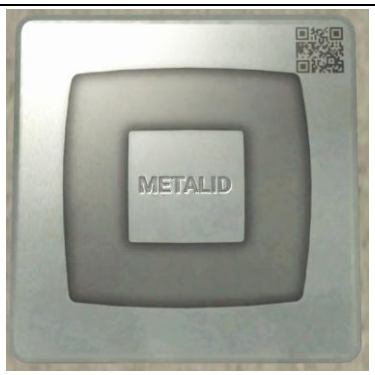
No.	Materials	Types of Materials
1.		Green Plastic Lid
2.		Metal Lid
3.		Green Plastic Base
4.		Metal Base

Table 1. Table of Materials

The hardware simulation utilizes four primary equipment categories to facilitate the sorting process *(Table 2).*

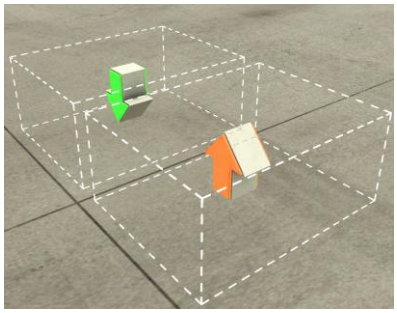
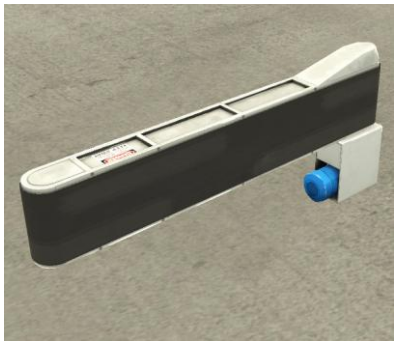
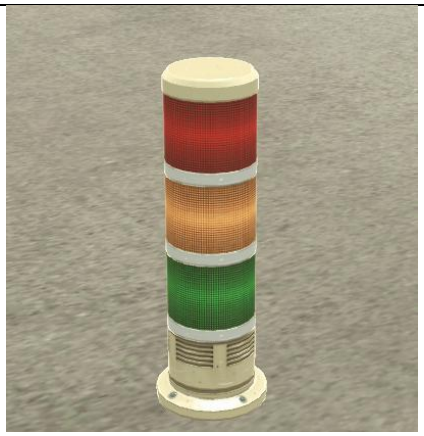
No.	Equipment	Description
1.		**Emitter and Remover :** - Emitter : Automatically generates the materials or parts (the lids and bases from *Table 1*). - Remover : Acts as a "sink" that deletes parts once they have been successfully sorted.
2.		**Main Control Panel :** This functions as the actual casing for the Human-Machine Interface (HMI).
3.		**Belt Conveyor :** The primary transport mechanism for the workpieces.
4.		**Pivot Arm Sorter :** The active sorting actuator.

Table 2. Table of Equipments

This table explains the control and sensing devices. These components are directly linked to your TIA Portal I/O tags.

No.	Components	Description
1.		**Start Button :** Sends a signal to the PLC to switch the system to a "Running" state.
2.		**Stop and Emergency Button :** For safety, stop all conveyor movement and sorting operations right away.
3.		**Reset Button :** Used specifically to clear data from all Counter Indicator.
4.		**Stack Light :** A visual signaling tower that indicates the current station status: Not Running:, Standby and Running.

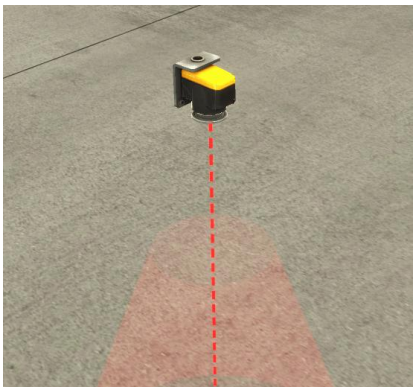
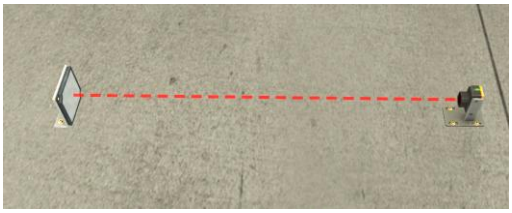
5.		**Counter Indicator :** A digital screen that shows data in real time. This project monitors four distinct metrics, Count total materials passed, Count of Metal Bases, Count of Green Plastic Bases and Count of all Lid materials.
6.		**Vision Sensor :** In order to tell between a Metal Lid, Plastic Lid, Metal Base, and Plastic Base, it recognizes the distinct "Digital ID" or characteristics of the components.
7.		**Retroreflective Sensor :** This sensor uses a light beam and a reflector. It serves a dual purpose Counting and Reset.

Table 3. Table of Components

C. Programming

PLC programming for the sorting station is carried out using Siemens TIA Portal, which provides a comprehensive environment for developing and testing ladder diagram (LAD) programs. Creating a new project and choosing a suitable PLC model to guarantee compatibility with the simulated automation system are the first steps in the programming process. After the project is established, the development of the control logic needed for the sorting process takes precedence.

All of the system's input and output signals are represented by PLC tags before the ladder logic is written. These include output signals that regulate conveyor motors and sorting systems, as well as Boolean input signals from sensors that identify the existence of workpieces. To enhance program readability and make the ladder diagram simpler to comprehend and troubleshoot during testing, symbolic naming is used for all tags.

The Ladder Diagram (LAD), which is frequently used in industrial automation because it provides a clear and graphical depiction of control logic, is then utilized to construct the control logic. The system's start and stop functions are managed by a self-holding (latching) circuit, which guarantees continued functioning until a stop condition is triggered. In the ladder networks, sensor signals are analyzed as Boolean criteria to decide when sorting actions and conveyor operations should take place.

Timers are used into the ladder program to add deliberate delays between activities in order to improve system control. By giving workpieces enough time to migrate into the proper locations before the subsequent action is carried out, these delays aid in process stabilization. Additionally, counters are utilized to tally the number of workpieces that the sorting station processes, allowing for basic output quantity and system performance monitoring.

The ladder diagram is arranged into logical, well-structured networks, each of which is in charge of a certain control function. The software may be readily expanded or changed in the future thanks to this structured programming method, which also increases clarity and decreases programming errors. All things considered, the PLC programming utilizing LAD in TIA Portal guarantees seamless connection with the Factory IO simulation environment and permits dependable operation of the sorting station.

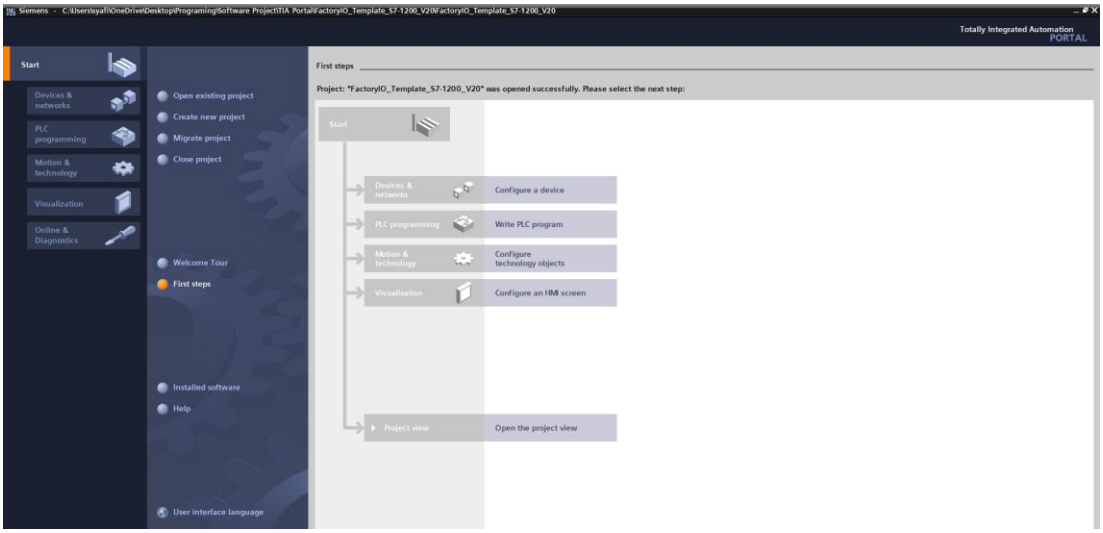


Figure 2. TIA Portal

- **Elements Used in Programming**

These elements form the fundamental electrical-style circuits used for starting, stopping, and controlling the system.

No.	Components	Description
1.		**Normally Open Contact (NO) :** This component allows power to flow only when the associated input or bit is "True" (active).
2.		**Normally Closed Contact (NC) :** When the bit is "False," this component permits power to flow; when the bit is "True," the circuit is broken.
3.		**Coil :** This is the output component.

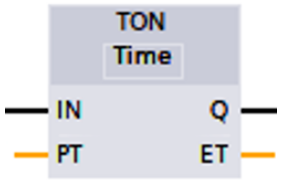
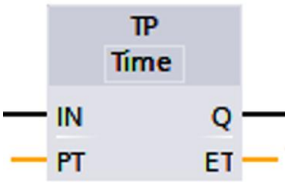
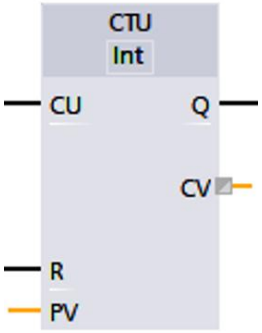
4.		**TON (Timer On-Delay) :** Timer that starts counting after got signal; Output turns on after counting.
5.		**TP (Timer Pulse) :** Timer that output turns on immediately after got signal for a fixed time.
6.		**CTU (Count Up) :** Every time its input is triggered, this component increases a value.

Table 4. Table of Elements

D. Simulation Communication

Siemens PLCSIM, a virtual PLC environment, is used to create communication between TIA Portal and Factory IO. The ladder diagram program created in TIA Portal can be run without a hardware PLC thanks to PLCSIM. This virtual PLC allows real-time data sharing with external simulation tools and functions like a real controller.

The PLC program is first compiled and downloaded into PLCSIM from the TIA Portal in this configuration. Factory IO is set up to use the Siemens S7-PLCSIM driver as its communication interface once the virtual PLC is operational. By enabling Factory IO to read input and output memory regions from the virtual PLC, this driver facilitates the flow of sensor signals and actuator commands between the two platforms.

Digital signals from input devices in Factory IO, including proximity or photoelectric sensors, are transmitted to the appropriate input addresses in PLCSIM. The TIA Portal's ladder diagram logic subsequently processes these signals. Conveyor motors and sorting mechanisms are examples of simulated actuators that are activated by the PLC sending output signals to Factory IO through PLCSIM based on the programmed conditions.

Real-time connectivity between TIA Portal, PLCSIM, and Factory IO enables instantaneous system behavior viewing. The PLC program can be tested, debugged, and validated with this configuration without the need for real hardware. An efficient virtual commissioning environment may be created by combining PLCSIM with Factory IO, which also helps guarantee that the control logic functions properly prior to real-world implementation.

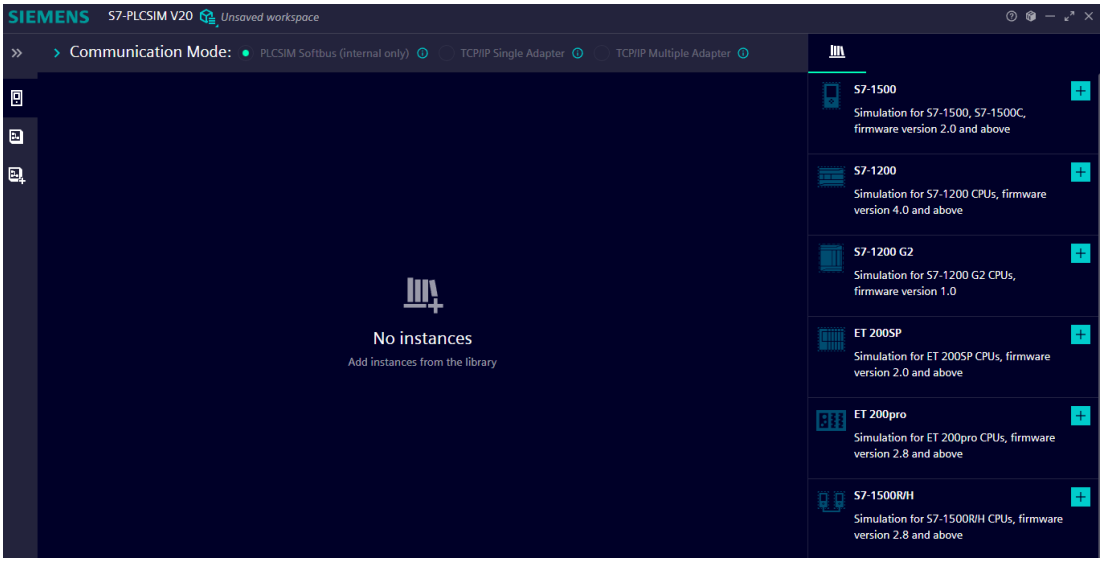


Figure 3. PLC SIM

RESULTS

The results of this project demonstrate the successful implementation and simulation of a PLC-controlled sorting station using TIA Portal, Ladder Diagram (LAD) programming, PLCSIM, and Factory IO. Programmed control logic and digital sensor inputs allowed the system to function as intended.

A. Hardware Simulation Result

The sorting station, which had several conveyor lines that represented various sorting paths, such as the metal base line, plastic base line, and plastic and metal lid line, was effectively emulated in Factory IO. Workpieces could be moved through the sorting process with ease since the conveyor system reacted appropriately to PLC output signals.

Workpieces could be precisely detected by sensors positioned along the conveyor. The ladder software used these sensor signals as Boolean inputs for decision-making once they were sent to the PLC via PLCSIM. During the simulation, the main control panel which had indicator lights and push buttons for start and stop operated as intended.



Figure 4.Sorting Station

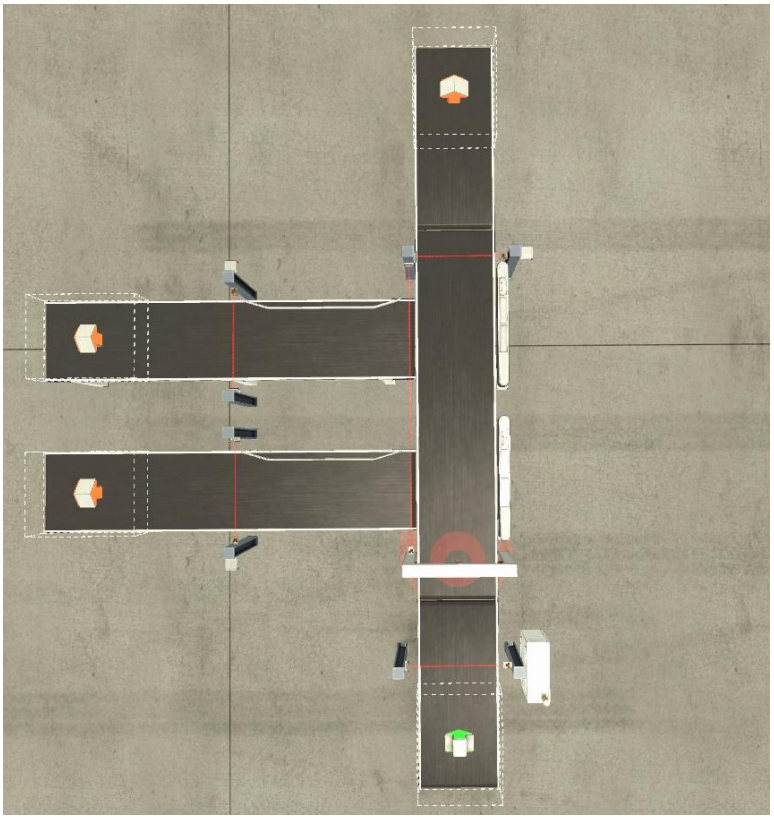


Figure 5.Sorting Station (Top View)



Figure 6.Main Control Panel

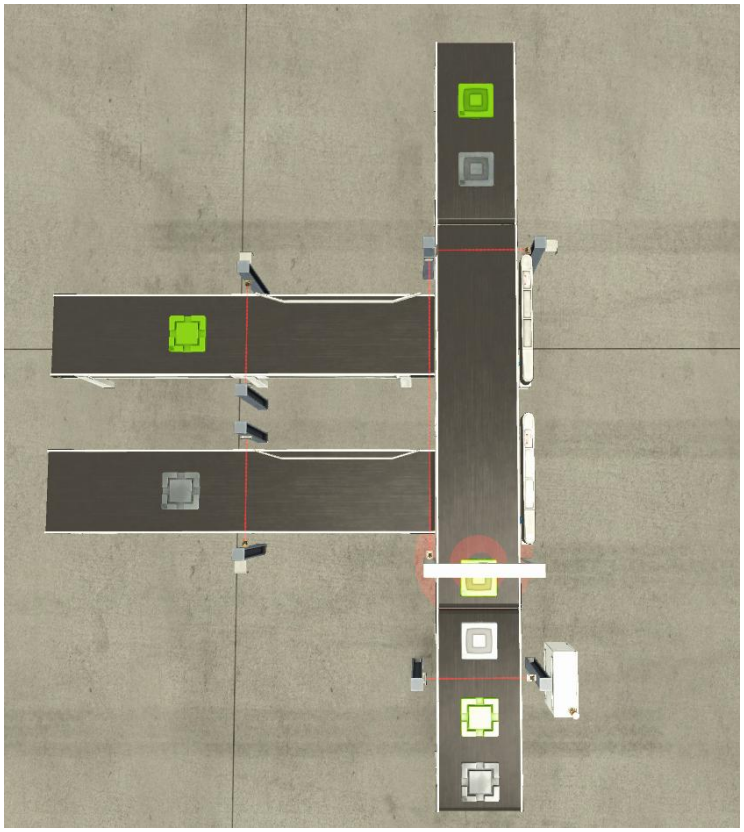


Figure 7. Separation Process

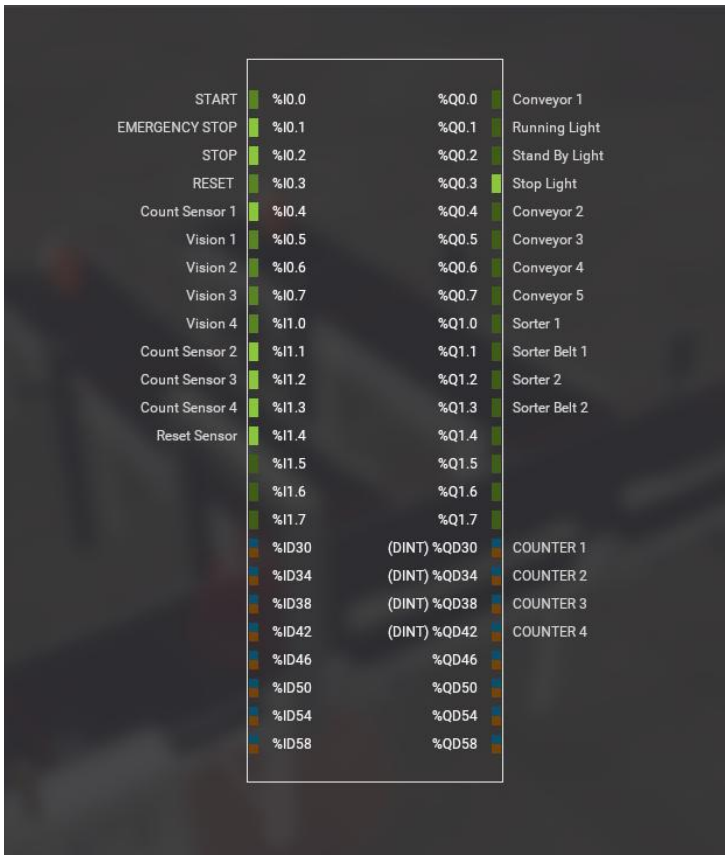


Figure 8. Driver Connections

B. Programming Result

When the PLC program created with LAD in the TIA Portal was downloaded to PLCSIM, it ran properly. Every PLC tag was identified and operated in accordance with its assigned symbolic address. The system was able to function in a clear and orderly manner because to the use of structured program blocks, which also enhanced readability.

After the start instruction was initiated, self-holding (latching) circuits guaranteed steady system functioning, while stop conditions promptly stopped the system when necessary. Accurate sorting behavior was made possible by conveyor control circuitry that reacted appropriately to sensor inputs.

The ladder software successfully implemented and tested timers and counters. The system was able to safely switch from standby mode to running mode thanks to the TON (On-Delay Timer), which offered a controlled delay during system startup. For particular movements, like actuator movement, the TP (Pulse Timer) produced fixed-duration signals.

The number of workpieces handled by the sorting station was precisely counted by the CTU (Up Counter). Reliable system throughput monitoring was demonstrated by counter numbers that were updated in real time and represented the accurate number of identified objects.

PLC tags

	Name	Tag table	Data type	Address	Retain	Acces...	Writa...	Visibl...
1	M1	MEMORY	Bool	%M10.0		☑	☑	☑
2	M2	MEMORY	Bool	%M20.0		☑	☑	☑
3	M3	MEMORY	Bool	%M30.0		☑	☑	☑
4	M4	MEMORY	Bool	%M40.0		☑	☑	☑
5	M5	MEMORY	Bool	%M50.0		☑	☑	☑
6	Start Button	INPUT	Bool	%I0.0		☑	☑	☑
7	Emergency Button	INPUT	Bool	%I0.1		☑	☑	☑
8	Count Sensor 1	INPUT	Bool	%I0.4		☑	☑	☑
9	Stop Button	INPUT	Bool	%I0.2		☑	☑	☑
10	Reset Button	INPUT	Bool	%I0.3		☑	☑	☑
11	Vision 1	INPUT	Bool	%I0.5		☑	☑	☑
12	Vision 2	INPUT	Bool	%I0.6		☑	☑	☑
13	Vision 3	INPUT	Bool	%I0.7		☑	☑	☑
14	Vision 4	INPUT	Bool	%I1.0		☑	☑	☑
15	Count Sensor 2	INPUT	Bool	%I1.1		☑	☑	☑
16	Count Sensor 3	INPUT	Bool	%I1.2		☑	☑	☑
17	Count Sensor 4	INPUT	Bool	%I1.3		☑	☑	☑
18	Reset Sensor	INPUT	Bool	%I1.4		☑	☑	☑
19	Running Light	OUTPUT	Bool	%Q0.1		☑	☑	☑
20	Stand By	OUTPUT	Bool	%Q0.2		☑	☑	☑
21	Stop Light	OUTPUT	Bool	%Q0.3		☑	☑	☑
22	Conveyor 1	OUTPUT	Bool	%Q0.0		☑	☑	☑
23	Conveyor 2	OUTPUT	Bool	%Q0.4		☑	☑	☑
24	Conveyor 3	OUTPUT	Bool	%Q0.5		☑	☑	☑
25	Conveyor 4	OUTPUT	Bool	%Q0.6		☑	☑	☑
26	Conveyor 5	OUTPUT	Bool	%Q0.7		☑	☑	☑
27	Sorter 1	OUTPUT	Bool	%Q1.0		☑	☑	☑
28	Sorter Belt 1	OUTPUT	Bool	%Q1.1		☑	☑	☑
29	Sorter 2	OUTPUT	Bool	%Q1.2		☑	☑	☑
30	Sorter Belt 2	OUTPUT	Bool	%Q1.3		☑	☑	☑
31	Counter 1	OUTPUT	DInt	%QD30		☑	☑	☑
32	Counter 2	OUTPUT	DInt	%QD34		☑	☑	☑
33	Counter 3	OUTPUT	DInt	%QD38		☑	☑	☑
34	Counter 4	OUTPUT	DInt	%QD42		☑	☑	☑
35	<Add new>					☑	☑	☑

Figure 9. PLC Tags

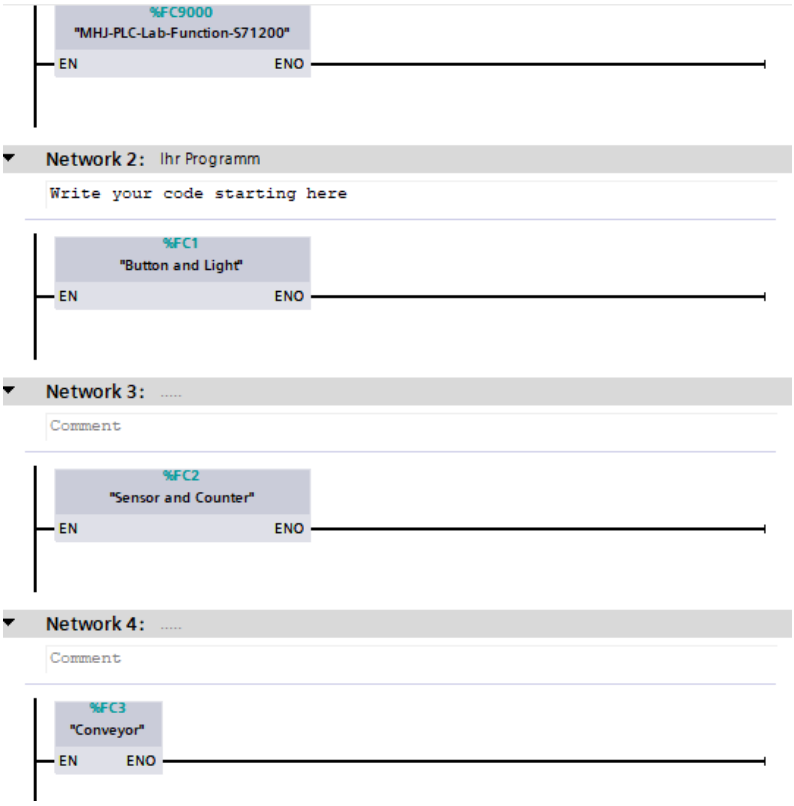


Figure 10. Main Block

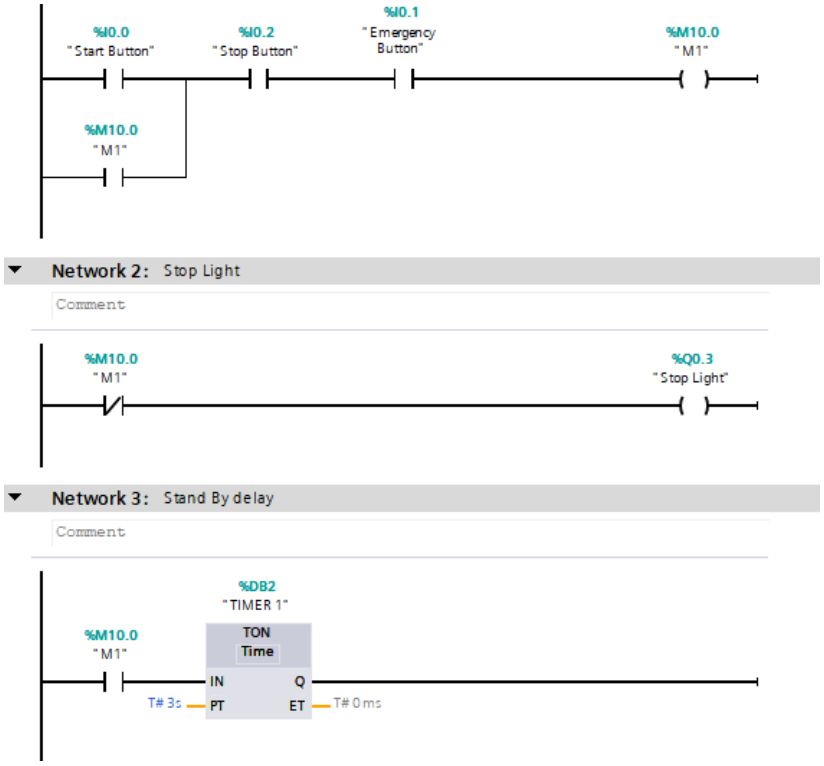


Figure 11. Button and Light Block i

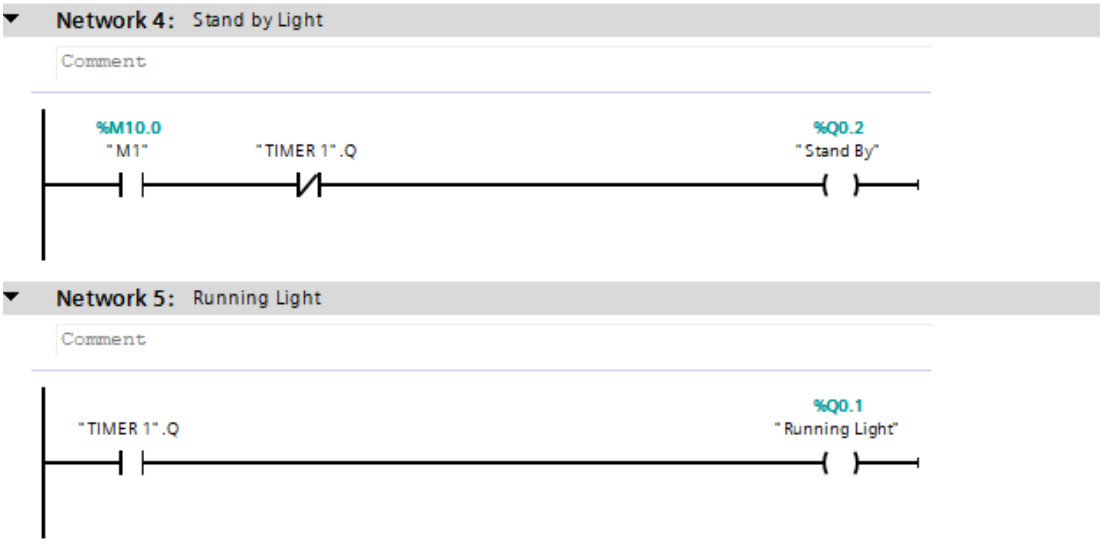


Figure 12. Button and Light Block ii

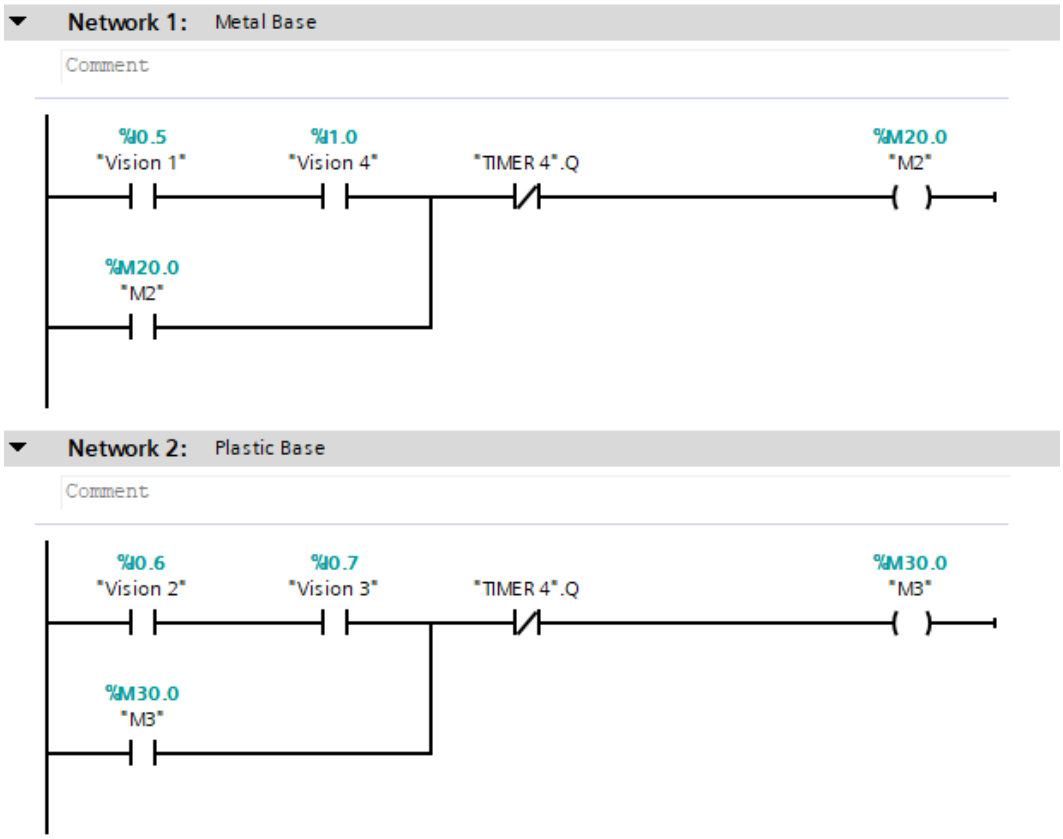


Figure 13. Sensor and Counter Block i

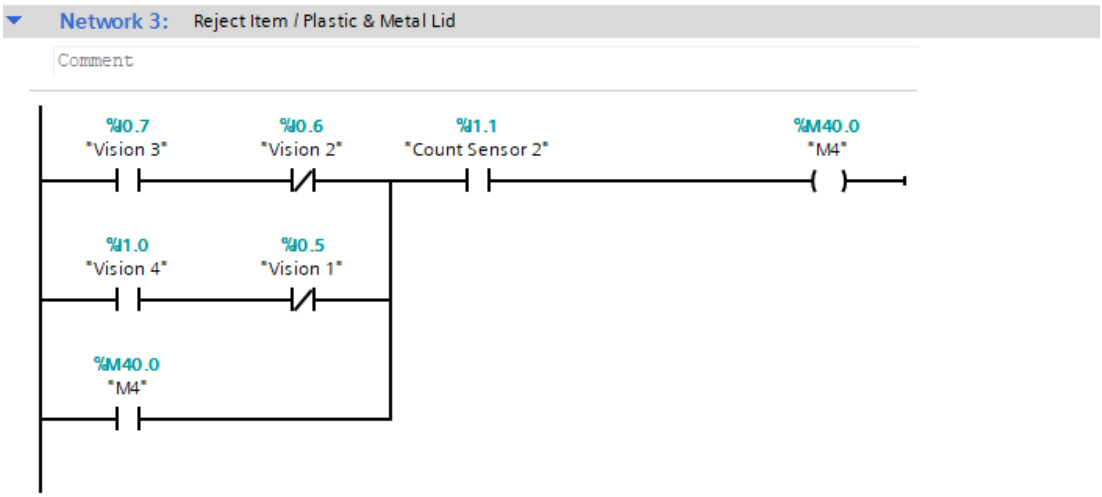


Figure 14. Sensor and Counter Block ii

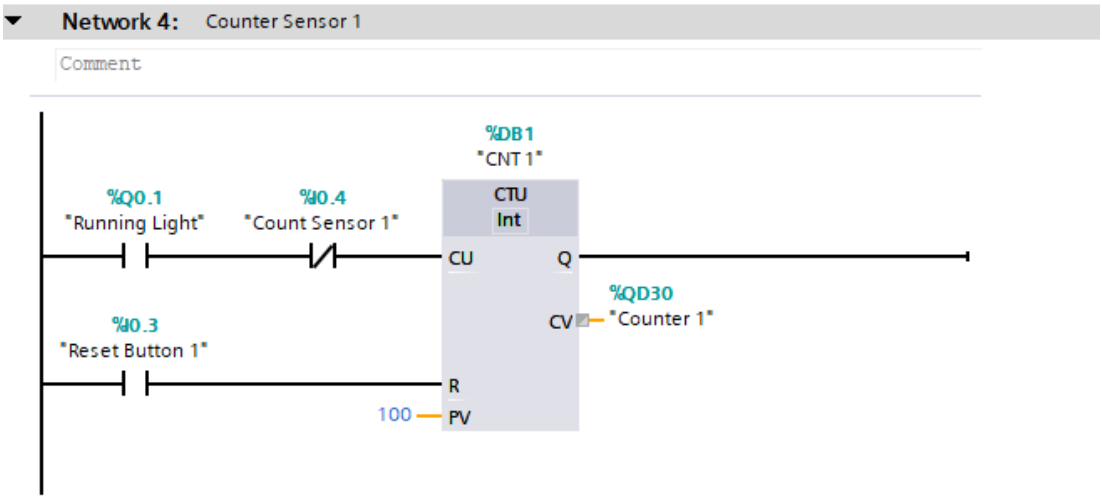


Figure 15. Sensor and Counter Block iii

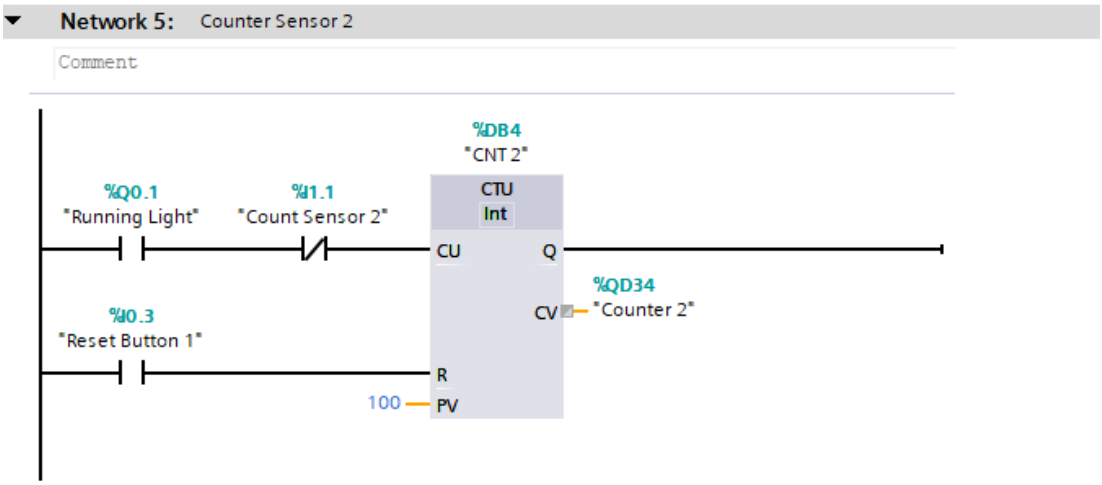


Figure 16. Sensor and Counter Block iv

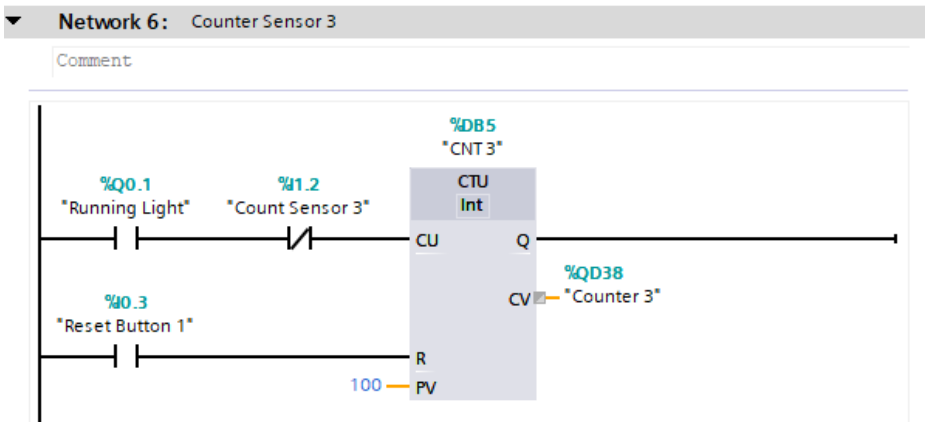


Figure 17. Sensor and Counter Block v

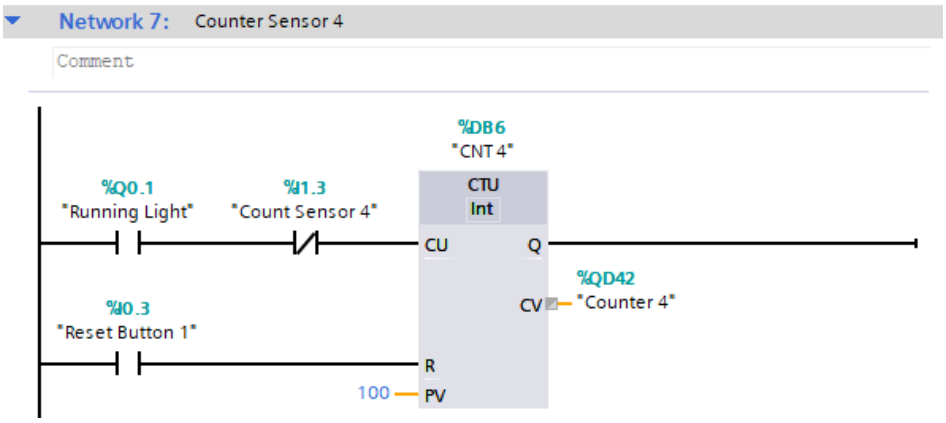


Figure 18. Sensor and Counter Block vi

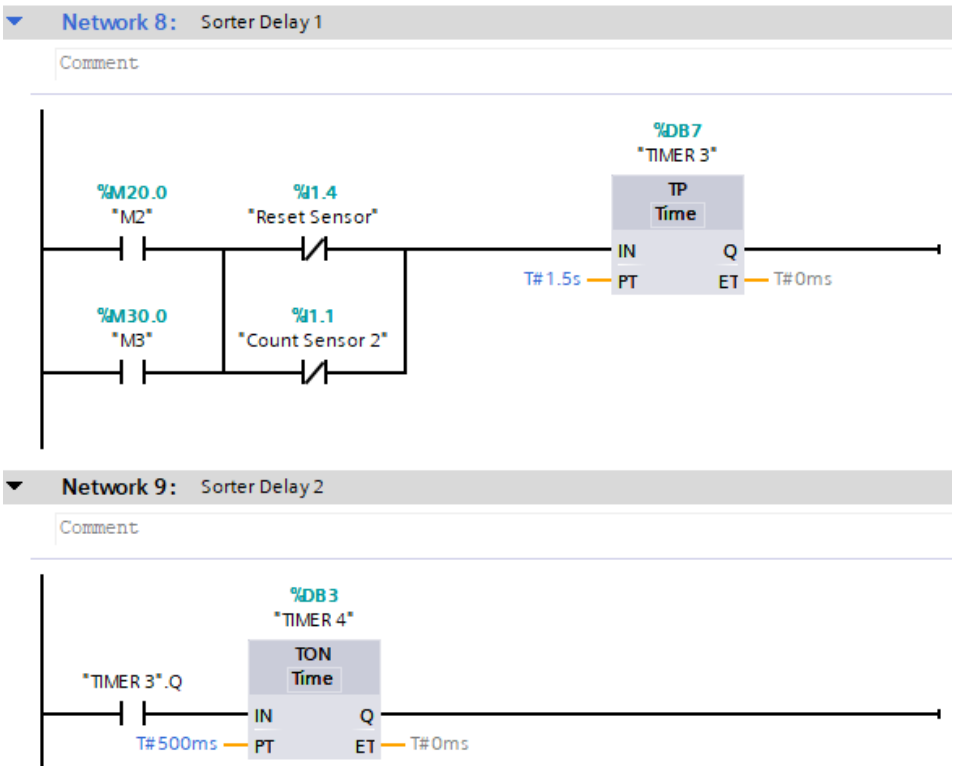


Figure 19. Sensor and Counter Block vii

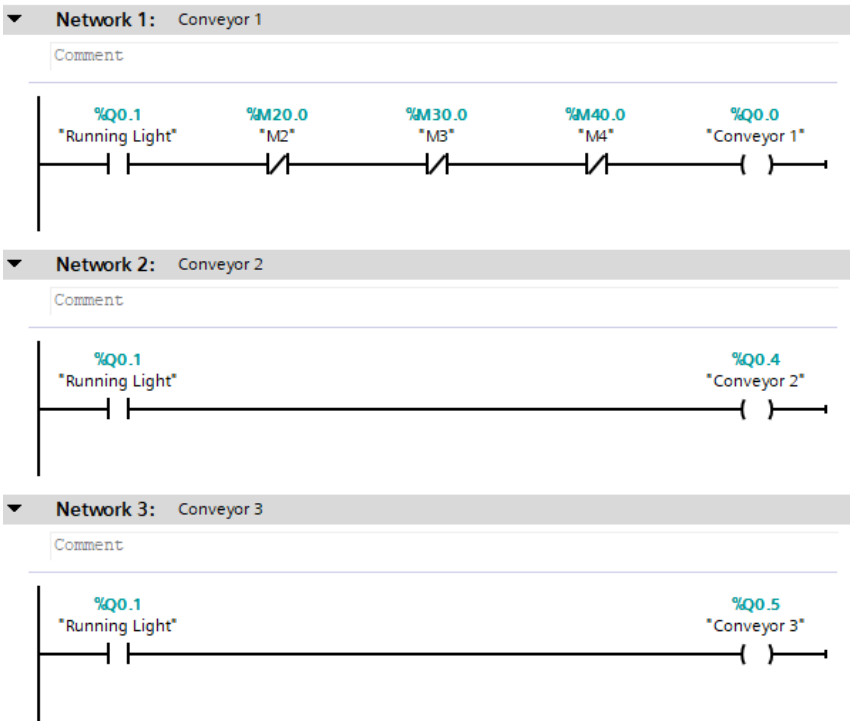


Figure 20. Conveyor Block i

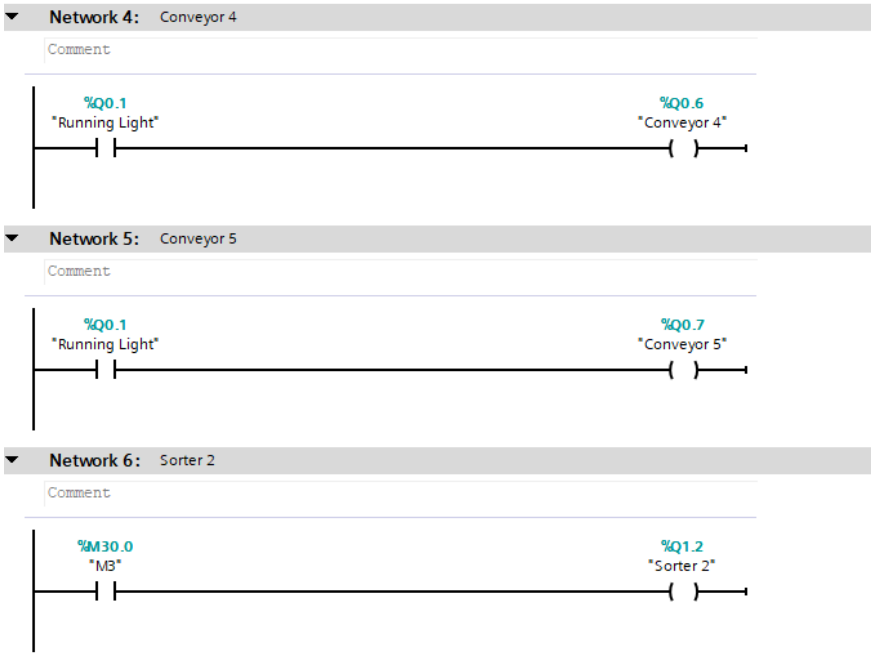


Figure 21. Conveyor Block ii

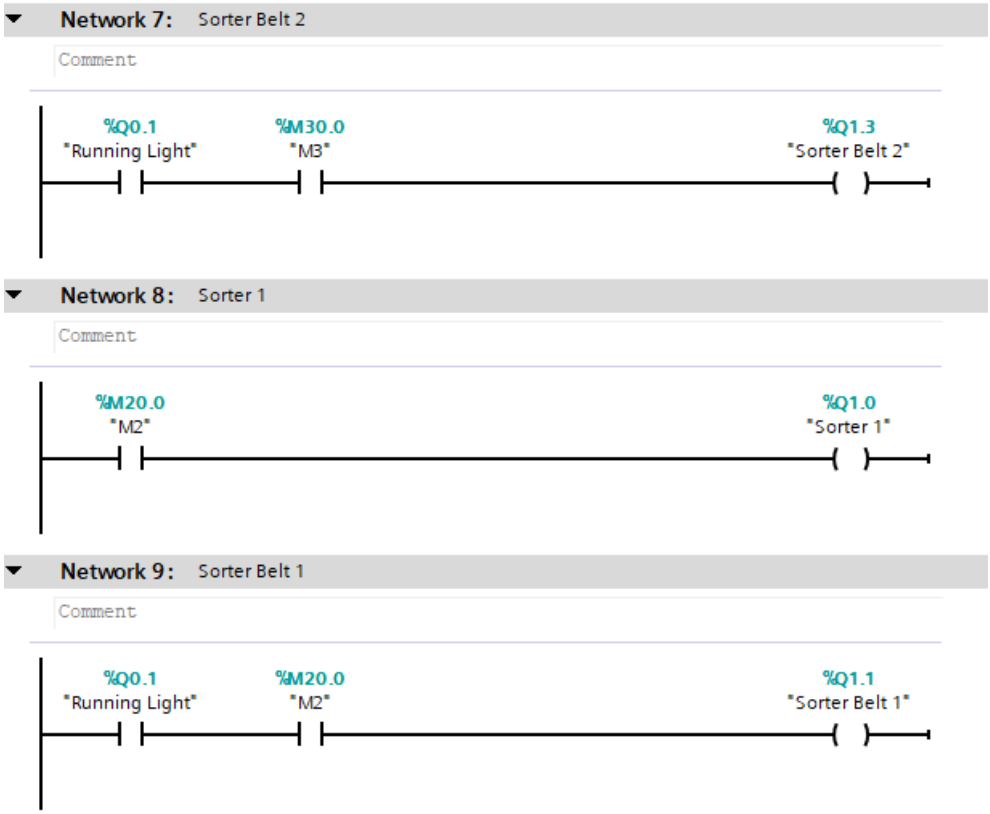


Figure 22. Conveyor Block iii

C. Simulation Communication Result

Successful communication was established between Factory IO, PLCSIM, and TIA Portal. The PLC program accurately accepted input signals from Factory IO sensors, and the simulated hardware was controlled by the PLC's output signals without any discernible delay. The virtual commissioning environment operated as anticipated, according to this real-time interaction.

Realistic testing and verification of the automation system were made possible by the use of PLCSIM, which removed the requirement for actual PLC hardware.

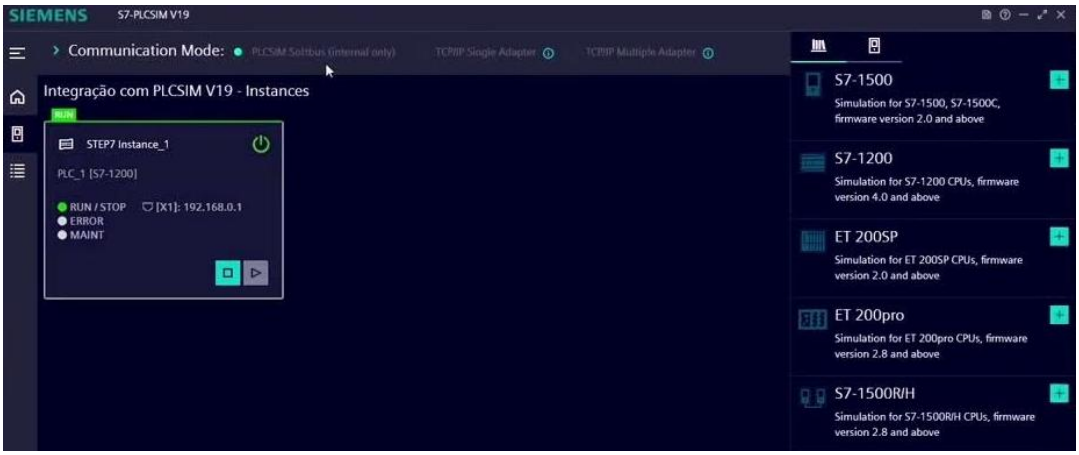


Figure 23. PLC SIM result

DISCUSSION

This project uses TIA Portal, Ladder Diagram (LAD), PLCSIM, and Factory IO to create an automated sorting system that illustrates the practical application of PLC programming ideas. The use of timers, the use of counters, and the choice not to employ RFID technology are the three primary discovery questions discovered in this study.

Answer Discovery Question

1. *How can timers be implemented and used in TIA Portal to control delays and sequencing in ladder diagram programming?*

Timers are crucial for managing the order and stability of automated systems. In this project, deliberate delays were introduced during system operation by using the TON (On-Delay Timer) within the ladder diagram. A predetermined time value was assigned to the timer, and Boolean logic criteria generated from sensor inputs and control signals were used to enable it.

By using timers, conveyors and sorting operations were prevented from starting right away, giving workpieces enough time to get to their proper locations. As a result, the simulation did not experience misalignment or unstable functioning. Timers in TIA Portal are easy to set up and very useful for controlling sequential control, particularly when working with continuous material flow systems, according to implementation and testing.

2. *How can counters be applied in TIA Portal to count and monitor the number of workpieces in an automated sorting system?*

To keep track of how many workpieces the sorting station processed, counters were installed. Every time a workpiece was picked up by a sensor, the count was increased using the CTU (Up Counter). To guarantee that every workpiece was counted just once, this required accurate edge detection.

In addition to enabling basic production tracking within the PLC software, the counter offered real-time feedback on system performance. This implementation revealed that TIA Portal counters are appropriate for applications needing process monitoring and quantity management. Additionally, the usage of counters showed how PLC programs may be expanded to incorporate data tracking and performance evaluation in addition to basic control logic.

3. *Why is RFID technology not implemented in this project, especially in relation to non-Boolean data handling and programming limitations?*

At first, RFID technology was thought to be useful for classifying and identifying workpieces according to extra details like material or type. However, due to knowledge and data processing constraints, RFID was not used in this project. In particular, non-Boolean data such as numerical or encoded information is usually provided by RFID devices, necessitating more sophisticated programming methods for data processing and interpretation.

On the other hand, the sensors employed in this experiment produced straightforward Boolean signals that are better suited for beginning PLC programming and simpler to incorporate into ladder logic. Furthermore, it was not feasible to implement within the parameters of this project due to a lack of documentation and a lack of experience with RFID data structures and programming techniques. Because of this, the project was able to concentrate on learning basic PLC principles, such as timers, counters, and Boolean logic, by excluding RFID.

CONCLUTION

This project successfully demonstrated the design and simulation of an automated sorting system using PLC programming with Ladder Diagram (LAD) in Siemens TIA Portal, integrated with PLCSIM and Factory IO. The system could be developed, tested, and validated without the need for actual hardware thanks to the virtual commissioning technique, which created a secure and affordable learning environment.

Through this experiment, it became evident how to use timers and counters in the TIA Portal. While counters allowed precise tracking of the quantity of workpieces processed by the sorting station, timers were successfully employed to manage delays and guarantee steady system performance. These programming components were effectively included into the ladder logic and are crucial to industrial automation.

Given the size of the task and the degree of programming complexity involved, the choice to forgo RFID technology was justifiable. The usage of basic Boolean sensors was more appropriate for meeting the project's learning objectives because RFID systems usually handle non-Boolean data and need for more sophisticated data processing techniques. This made it possible to concentrate more on system integration and basic PLC programming techniques.

All things considered, the project met its goals and offered useful hands-on training in PLC programming, industrial simulation, and system integration. For more complex automation projects including intricate data management and practical industrial applications, the expertise acquired from this work provides a solid basis.

Future Work

Even though this assignment's goals were successfully met, there are a number of enhancements and additions that could be made in the future to increase the system's functionality and industrial relevance. Using RFID technology to identify workpieces is one potential improvement. The system may handle non-Boolean data and carry out more sophisticated sorting based on encoded information such material type, batch number, or destination by integrating RFID readers and tags.

Expanding the PLC program to incorporate data processing and sophisticated logic capabilities, like data blocks, numerical value comparison, and structured programming techniques, is another possible improvement. This would enhance the system's capacity to manage intricate decision-making procedures and get it ready for practical industrial uses.

Another way to expand the system would be to incorporate Human-Machine Interface (HMI) features. Operators may check production counts, monitor system status, change timer values, and more effectively manage system operation with the help of an HMI. This would represent standard procedures in contemporary automated manufacturing systems and enhance usability.

Lastly, the approved PLC program may be transferred from the simulation environment to actual hardware, such as actuators, sensors, and PLCs. This would make it possible to test in real-world settings and offer more information on hardware limitations, safety issues, and system dependability.

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